

Книга: **MicroPython** за **VK_RA4M2** чрез демонстрационни примери

Цел

46

.py

VK_RA4M2

1.

2.

3.

Как да се ползва книгата

examples

1.

2.

3. .py

4.

5.

DO_WRITE DO_BOOTLOADER DO_LIGHTSLEEP

DO_DEEPSLEEP SET_DEMO_TIME

Ресурсна карта на **VK_RA4M2**

- 1 брой LED1 = P204
- 1 брой SW1 = P400
- USBDP = P914 USBDM = P915 USB_VBUS = P407
- 13 броя P000 P001 P002 P003 P004 P005 P006 P007 P008 P013 P014 P015 P500
- 3 броя ADC.CORE_TEMP ADC.CORE_VREF ADC.VREF

- 2 броя P014 P015
- 14 броя P107 P106 P105 P104 P113 P114 P112 P115 P608 P409 P408 P600 P304 P303
- 4 инстанции UART(0) UART(2) UART(7) UART(9)
- 2 инстанции I2C(0)=P400/P401 I2C(1)=P100/P101
- 2 инстанции I2CTarget(0)=P400/P401 I2CTarget(1)=P100/P101
- 1 канал CS=P103 SCK=P102 MISO=P100 MOSI=P101
-
-
- 12 броя P205 P206 P407 P408 P409 P410 P411 P412 P413 P414 P415 P708
- P207 = TSCAP
- 2 броя Timer(1) Timer(2)
- Timer(-1)
- 1 брой
- /flash
- renesas.Flash
- dataflash 8 KB
- SPI 1
- Timer(-1)
- SW1 P400 I2C(0).SCL
- LED1 0 1
- Pin.PULL_UP Pin.PULL_DOWN
- lightsleep deepsleep

Матрица за пълност спрямо плана

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- Логически оператори examples/20_language_basics/03_operators_conditions_loops.py
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- switch-case в MicroPython липсва
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- Динамична 7-сегментна индикация
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- Декомпозиция на 2 FSM `examples/31_case_study/02_two_fsms.py`
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Част I: Първи стъпки с борда и с **MicroPython**

Глава 1. Първа програма

VK_RA4M2

```
from machine import Pin
import time

led = Pin("LED1", Pin.OUT, value=1)

for blink_index in range(3):
    led.value(0)
    time.sleep_ms(300)
    led.value(1)
    time.sleep_ms(300)
```

- Pin
 - for
 - sleep_ms
 - LED1
- examples/20_language_basics/01_sample_program.py

Глава 2. Първи асинхронни примери

```
import asyncio
from machine import Pin

led = Pin("LED1", Pin.OUT, value=1)

async def blink_task():
    while True:
        led.value(0)
        await asyncio.sleep_ms(500)
        led.value(1)
        await asyncio.sleep_ms(500)

async def count_seconds_task():
    seconds = 0
    while True:
        seconds += 1
        print("Изминали секунди:", seconds)
        await asyncio.sleep(1)
```

- asyncio
 -
- examples/01_blink_async.py

- `examples/02_two_tasks.py`

Част II: Езикови основи на **MicroPython**

Глава 3. Променливи и константи

```
from micropython import const

BOARD_NAME = "VK_RA4M2"
LED_ALIAS = "LED1"
BLINK_COUNT = const(3)
BLINK_DELAY_MS = const(200)
uses_asyncio = True
```

-

- `const()`

- `examples/20_language_basics/02_variables_and_constants.py`

Глава 4. Оператори, условия и цикли

```
left_value = 7
right_value = 3

sum_value = left_value + right_value
logic_result = (left_value > right_value) and (left_value != right_value)

if left_value < right_value:
    print("По-малка")
elif left_value == right_value:
    print("Равна")
else:
    print("По-голяма")
```

```

for index in range(3):
    print(index)

counter = 0
while counter < 3:
    print(counter)
    counter += 1

attempt = 0
while True:
    print(attempt)
    attempt += 1
    if attempt >= 2:
        break

```

- goto
 - switch-case
 - if/elif/else break continue return
 - do-while while True break
- examples/20_language_basics/03_operators_conditions_loops.py

Глава 5. Enums в MicroPython

```

from micropython import const

STATE_OFF = const(0)
STATE_ON = const(1)
STATE_BLINK = const(2)

```

- -
- examples/20_language_basics/04_enums.py

Глава 6. Массивы и таблицы

```

from array import array

adc_samples = [1200, 1215, 1198, 1222]
led_table = [["LED1", "P204"], ["SW1", "P400"]]
pwm_duty_values = array("H", [0, 16384, 32768, 49152, 65535])
raw_bytes = bytearray([0x10, 0x20, 0x30, 0x40])

```

- list
- list
- array
- bytearray

- examples/20_language_basics/05_arrays.py

Глава 7. • структуры в **MicroPython**

struct

- dict
- namedtuple
-
-

```

from collections import namedtuple

```

```

SensorRecord = namedtuple("SensorRecord", ("name", "pin", "unit"))
light_sensor = {"name": "Light", "adc_pin": "P000", "limits": {"dark": 800, "bright": 3000}}
devices = [{"name": "LED1", "pin": "P204", "kind": "output"}]

```

- dict
- dict

- examples/20_language_basics/06_structures.py

Глава 8. ,казатели и еквивалентите им в **MicroPython**

```
adc_list = [100, 200, 300]
alias_list = adc_list
alias_list[1] = 999

tx_buffer = bytearray([1, 2, 3, 4])
tx_view = memoryview(tx_buffer)
tx_view[2] = 77
```

- -
 - `memoryview`
 - `machine.mem8 mem16 mem32`
-
- `examples/20_language_basics/07_pointers_equivalents.py`
 - `examples/28_machine_misc/02_machine_mem_and_bootloader.py`

Глава 9. **f** ункции, **callbacks** и таблици от „ункции

```
def add_values(left_value, right_value):
    return left_value + right_value

def run_callback(callback, value):
    callback(value)

def square(value):
    return value * value

operation_table = [square]
```

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- `examples/20_language_basics/08_functions.py`

Част III: Ци „рови и аналогови входи и изходи

Глава 10. Ци „ров изход

```
from machine import Pin
import time

led = Pin("LED1", Pin.OUT, value=1)
led.value(0)
time.sleep_ms(700)
led.value(1)
```

- 0
- 1
-
-

- `examples/21_digital_io/01_digital_output.py`
- `examples/01_blink_async.py`

Глава 11. Ци „ров вход

```
from machine import Pin

button = Pin("SW1", Pin.IN, Pin.PULL_UP)
print(button.value())
```

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- PULL_UP
- PULL_DOWN

- examples/21_digital_io/02_digital_input.py

Глава 12. ...налогов вход и ADC

```
from machine import ADC

temperature_adc = ADC(ADC.CORE_TEMP)
vref_adc = ADC(ADC.CORE_VREF)

print(temperature_adc.read())
print(vref_adc.read())
```

-
-
- read_u16()
-

P000 P001 P002 P003 P004 P005 P006 P007 P008 P013 P014 P015 P500

- examples/22_analog/01_analog_input_adc.py

Глава 13. PWM

```
from machine import Pin, PWM

pwm = PWM(Pin("P107"), freq=1000, duty=50)

pwm_a = PWM(Pin("P107"), freq=1000, duty=25)
```

```
pwm_b = PWM(Pin("P106"), freq=1000, duty=75)
pwm_a.freq(2000)
```

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- examples/22_analog/02_pwm_output.py
- examples/03_pwm_ab_same_channel.py

Част IV: таџмери, прекъсвания и време

Глава 14. ^акъснения, хардуерни таџмери и Timer(-1)

```
import time
time.sleep_ms(300)
```

```
from machine import Timer

timer = Timer(1)
timer.init(freq=4, callback=timer_callback, hard=False)
```

```
soft_timer = Timer(-1)
soft_timer.init(freq=4, callback=soft_timer_callback, hard=False)
```

- Timer(1) Timer(2)
- Timer(-1)
- asyncio

- examples/23_timing/01_delays_and_timers.py
- examples/23_timing/04_software_timer_minus1.py
- examples/23_timing/05_timer_compare_hw_sw.py

Глава 15. Прекъсвания

```
from machine import Pin, disable_irq, enable_irq

button = Pin("SW1", Pin.IN, Pin.PULL_UP)

def button_handler(pin_object):
    print("IRQ")

button.irq(handler=button_handler, trigger=Pin.IRQ_FALLING, hard=False)
```

```
irq_state = disable_irq()
snapshot = irq_state_data["count"]
enable_irq(irq_state)
```

- `examples/23_timing/02_interrupts.py`
- `examples/28_machine_misc/02_machine_mem_and_bootloader.py`

Глава 16. %ебаунс на бутон

```
if current_value != last_stable_value:
    now_ms = time.ticks_ms()
    if time.ticks_diff(now_ms, last_change_ms) >= 40:
        last_change_ms = now_ms
        last_stable_value = current_value

    uasyncio

if stable_value == 0 and stable_count == 4:
    async_presses += 1
```

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-

- examples/23_timing/03_button_debounce.py

Глава 17. RTC

```
from machine import RTC

rtc = RTC()
print(rtc.info())
print(rtc.calibration())
print(rtc.datetime())
```

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-

- examples/23_timing/06_rtc_basic.py

Част V: • ериџни интер „ еџси

Глава 18. I2C master и runtime избор на пинове

```
from machine import I2C

i2c = I2C(1, freq=100000)
print(i2c.scan())
```

```
from machine import I2C, Pin

i2c = I2C(1, freq=400000, scl=Pin("P100"), sda=Pin("P101"))
```

- I2C(0) P400/P401
- P400 SW1

- examples/24_i2c/01_i2c_master_basic.py
- examples/24_i2c/02_i2c_runtime_pins.py

Глава 19. SoftI2C и I2CTarget

```
from machine import SoftI2C, Pin
```

```
soft_i2c = SoftI2C(scl=Pin("P105", Pin.OPEN_DRAIN), sda=Pin("P104", Pin.OPEN_DRAIN), fr  
eq=100000)
```

```
from machine import I2CTarget, Pin
```

```
memory_buffer = bytearray(range(16))
```

```
target = I2CTarget(1, addr=0x42, mem=memory_buffer, mem_addrsize=8, scl=Pin("P100"), sd  
a=Pin("P101"))
```

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- examples/24_i2c/03_softi2c_basic.py
- examples/24_i2c/04_i2ctarget_memory.py

Глава 20. Hardware SPI и SoftSPI

```
from machine import SPI, Pin
```

```
spi = SPI(0, baudrate=500000, polarity=0, phase=0, bits=8, sck=Pin("P102"), mosi=Pin("P  
101"), miso=Pin("P100"), cs=Pin("P103"))
```

```
from machine import SoftSPI, Pin
```

```
soft_spi = SoftSPI(baudrate=100000, polarity=0, phase=0, sck=Pin("P105"), mosi=Pin("P10  
4"), miso=Pin("P106"))
```

- examples/25_spi/01_spi_basic.py
- examples/25_spi/02_softspi_basic.py

Част VI: **TouchPad**, памет и системни „услуги“

Глава 21. **TouchPad**: базово четене

```
from machine import Pin, TouchPad

touch = TouchPad(Pin("P205"))
touch.config(500)
print(touch.read())
print(touch.value())
print(touch.read_value())
```

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- examples/26_touchpad/01_touchpad_basic.py

Глава 22. **TouchPad**: диагностика, **cached API** и асинхронно следене

```
TouchPad.sample_rate(20)
touch.start()
TouchPad.service()
print(touch.ready(), touch.read_cached(), touch.value_cached())
print(touch.diagnose(8))
```

```
TouchPad.sample_rate(50)

async def touch_task():
    while True:
```



```

if tp.ready():
    print(tp.read_cached(), tp.value_cached(), tp.age_ms())
await asyncio.sleep_ms(20)

```

- examples/26_touchpad/02_touchpad_diagnostics.py
- examples/04_touchpad_async.py

Глава 23. renesas.Flash, /flash и dataflash

renesas.Flash

```

import os
import renesas

flash = renesas.Flash()
sector = bytearray(16)
flash.readblocks(0, sector)
print(os.listdir("/flash"))

```

dataflash

```

import dataflash

print(dataflash.size())
print(dataflash.block_size())
print(dataflash.write_size())
print(dataflash.read(0, 16))

```

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- examples/27_storage/01_flash_blockdev.py
- examples/27_storage/02_dataflash_basic.py

Глава 24. machine модулът и порт-специфично поведение

```

import machine

uid = machine.unique_id()
print(machine.freq())
print(machine.reset_cause())
machine.info()

irq_state = machine.disable_irq()
machine.enable_irq(irq_state)
print(hasattr(machine, "mem32"))

machine.sleep(20)

if DO_LIGHTSLEEP:
    machine.lightsleep(100)

```

- machine.info
- machine.freq
- machine.unique_id
- machine.reset_cause
- disable_irq enable_irq
- mem8 mem16 mem32
- machine.bootloader
- sleep lightsleep deepsleep

- examples/28_machine_misc/01_machine_info_and_id.py
- examples/28_machine_misc/02_machine_mem_and_bootloader.py
- examples/28_machine_misc/03_sleep_modes.py

Част VII: Индикация и входни матрици

Глава 25. Динамична 7-сегментна индикация и матрично сканиране

```

segment_pins = [Pin(name, Pin.OUT) for name in ("P105", "P104", "P113", "P114", "P115",
"P304", "P303")]
digit_pins = [Pin(name, Pin.OUT) for name in ("P608", "P409")]

for digit_index, symbol in enumerate(("1", "2")):
    for digit_pin in digit_pins:
        digit_pin.off()
    for segment_pin, level in zip(segment_pins, patterns[symbol]):
        segment_pin.value(level)
    digit_pins[digit_index].on()
    time.sleep_ms(5)

for row_index, row_pin in enumerate(row_pins):
    for reset_row in row_pins:
        reset_row.off()
    row_pin.on()
    for col_index, col_pin in enumerate(col_pins):
        if col_pin.value() == 0:
            pressed_keys.append((row_index, col_index))

```

- examples/29_display_scan/01_dynamic_7seg_and_keypad.py

Част VIII: FSM като начин на мислене

Глава 26. Минимална FSM и таблична FSM

```

STATE_OFF = "OFF"
STATE_ON = "ON"
EVENT_BUTTON = "BUTTON"

def handle_event(current_state, event_name):
    if current_state == STATE_OFF and event_name == EVENT_BUTTON:
        return STATE_ON
    if current_state == STATE_ON and event_name == EVENT_BUTTON:

```

```
        return STATE_OFF
    return current_state
```

```
transition_table = {
    ("OFF", "BUTTON"): "ON",
    ("ON", "BUTTON"): "OFF",
}
```

```
OFF --BUTTON--> ON
ON  --BUTTON--> OFF
```

- examples/30_fsm/01_fsm_lamp_button.py
- examples/30_fsm/02_fsm_table.py

Глава 27. FSM с timeout

```
transition_table = {
    ("OFF", "BUTTON"): "ON",
    ("ON", "TIMEOUT"): "OFF",
    ("ON", "BUTTON"): "OFF",
}
```

- examples/30_fsm/03_fsm_timeout.py

Глава 28. Conditions vs Events

```
light_level = 600
dark_threshold = 800
event_name = "BUTTON"
condition_is_dark = light_level < dark_threshold
```

```
if condition_is_dark and event_name == "BUTTON":  
    print("Лампата може да се включи")
```

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-

- `examples/30_fsm/04_conditions_vs_events.py`

Глава 29. Guard „ункции

```
def is_dark(data):  
    return data["light"] < 800  
  
def has_motion(data):  
    return data["pir"] is True  
  
if state == "OFF" and event_name == "MOTION" and is_dark(context) and has_motion(context):  
    state = "ON"
```

- `examples/30_fsm/05_guard_functions.py`

Глава 30. Приоритети и **event queue**

```
pending_events = ["BUTTON", "ERROR", "TIMEOUT"]  
priority_order = {"ERROR": 0, "BUTTON": 1, "TIMEOUT": 2}  
pending_events.sort(key=lambda event_name: priority_order[event_name])
```

```
event_queue = []  
event_queue.append(("BUTTON", {"source": "SW1"}))  
event_queue.append(("PIR", {"source": "pir_sensor"}))  
  
while event_queue:
```

```
event_name, payload = event_queue.pop(0)
print(event_name, payload)
```

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- examples/30_fsm/06_priorities.py
- examples/30_fsm/07_event_queue.py

Глава 31. Tickless timers и sleep идея

```
tasks = [("lamp_timeout", 120), ("fan_timeout", 450), ("display_refresh", 40)]
next_deadline_ms = min(deadline for _, deadline in tasks)
machine.sleep(sleep_ms)
ready_tasks = [task_name for task_name, deadline in tasks if deadline <= elapsed_ms]
```

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- examples/30_fsm/08_tickless_sleep.py
- examples/28_machine_misc/03_sleep_modes.py

Част IX: Реален пример

Глава 32. , правление на осветление и вентилация за баня

```
sensors = {
    "pir": True,
    "door_open": False,
```

```

    "light_level": 300,
    "humidity": 82,
}

outputs = {"lamp": False, "fan": False}

if sensors["pir"] and sensors["light_level"] < 800:
    outputs["lamp"] = True

if sensors["humidity"] > 75 or sensors["door_open"]:
    outputs["fan"] = True

```

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• examples/31_case_study/01_bathroom_control.py

Глава 33. %окомпозиция на две **FSM**

```

lamp_state = "OFF"
fan_state = "OFF"

for event_name in event_sequence:
    if event_name == "BUTTON":
        lamp_state = "ON" if lamp_state == "OFF" else "OFF"
    if event_name == "HUMIDITY_HIGH":
        fan_state = "ON"

```

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- `examples/31_case_study/02_two_fsms.py`

Каталог на всички примери

- `examples/01_blink_async.py` LED1
- `examples/02_two_tasks.py` asyncio
- `examples/03_pwm_ab_same_channel.py`
- `examples/04_touchpad_async.py`
- `examples/20_language_basics/01_sample_program.py`
- `examples/20_language_basics/02_variables_and_constants.py`
- `examples/20_language_basics/03_operators_conditions_loops.py`
- `examples/20_language_basics/04_enums.py` const()
- `examples/20_language_basics/05_arrays.py` array bytearray
- `examples/20_language_basics/06_structures.py` dict namedtuple
- `examples/20_language_basics/07_pointers_equivalents.py` memoryview
machine.mem32
- `examples/20_language_basics/08_functions.py`
- `examples/21_digital_io/01_digital_output.py` LED1
- `examples/21_digital_io/02_digital_input.py` SW1
- `examples/22_analog/01_analog_input_adc.py`
- `examples/22_analog/02_pwm_output.py`
- `examples/23_timing/01_delays_and_timers.py` uasyncio
- `examples/23_timing/02_interrupts.py`
- `examples/23_timing/03_button_debounce.py` uasyncio
- `examples/23_timing/04_software_timer_minus1.py` Timer(-1)
- `examples/23_timing/05_timer_compare_hw_sw.py` Timer(1) Timer(-1)
- `examples/23_timing/06_rtc_basic.py`
- `examples/24_i2c/01_i2c_master_basic.py`
- `examples/24_i2c/02_i2c_runtime_pins.py`
- `examples/24_i2c/03_softi2c_basic.py`
- `examples/24_i2c/04_i2ctarget_memory.py`
- `examples/25_spi/01_spi_basic.py`
- `examples/25_spi/02_softspi_basic.py`
- `examples/26_touchpad/01_touchpad_basic.py`

- examples/26_touchpad/02_touchpad_diagnostics.py
- examples/27_storage/01_flash_blockdev.py renesas.Flash /flash
- examples/27_storage/02_dataflash_basic.py dataflash
- examples/28_machine_misc/01_machine_info_and_id.py machine.info freq unique_id
- examples/28_machine_misc/02_machine_mem_and_bootloader.py disable_irq enable_irq mem32 bootloader
- examples/28_machine_misc/03_sleep_modes.py sleep lightsleep deepsleep
- examples/29_display_scan/01_dynamic_7seg_and_keypad.py
- examples/30_fsm/01_fsm_lamp_button.py
- examples/30_fsm/02_fsm_table.py
- examples/30_fsm/03_fsm_timeout.py
- examples/30_fsm/04_conditions_vs_events.py
- examples/30_fsm/05_guard_functions.py
- examples/30_fsm/06_priorities.py
- examples/30_fsm/07_event_queue.py
- examples/30_fsm/08_tickless_sleep.py
- examples/31_case_study/01_bathroom_control.py
- examples/31_case_study/02_two_fsms.py

Какво о κ е не е потвърдено на реален борд

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- Flash dataflash
- lightsleep deepsleep

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